again leading us back to the big bang. How about the partially melted ice cubes you saw at 10:30 p.m.? Now that we are trusting memories and records, you remember that just before 10 p.m. the bartender put fully formed ice cubes in your glass. He got the ice cubes from a freezer, which was designed by a clever engineer and fabricated by talented machinists, all of whom are capable of creating something of such high order because they themselves are highly ordered life forms. And again, we can sequentially trace their order back to the highly ordered origin of the universe.

The Critical Input

The revelation we've come to is that we can trust our memories of a past with lower, not higher, entropy only if the big bang—the process, event, or happening that brought the universe into existence—started off the universe in an extraordinarily special, highly ordered state of low entropy. Without that critical input, our earlier realization that entropy should increase toward both the future and the past from any given moment would lead us to conclude that all the order we see arose from a chance fluctuation from an ordinary disordered state of high entropy, a conclusion, as we've seen, that undermines the very reasoning on which it's based. But by including the unlikely, low-entropy starting point of the universe in our analysis, we now see that the correct conclusion is that entropy increases toward the future, since probabilistic reasoning operates fully and without constraint in that direction; but entropy does not increase toward the past, since *that* use of probability would run afoul of our new proviso that the universe began in a state of low, not high, entropy.[17] Thus, conditions at the birth of the universe are critical to directing time's arrow. *The future is indeed the direction of increasing entropy. The arrow of time—the fact that things start like* this *and end like* that *but never start like* that *and end like* this*—began its flight in the highly* ordered, low-entropy state of the universe at its inception.[18]

The Remaining Puzzle

That the early universe set the direction of time's arrow is a wonderful and satisfying conclusion, but we are not done. A huge puzzle remains.